MIL-C-43480B

24 February 1977
SUPERSEDING
MIL-C-43480A(GL)
18 March 1971

MILITARY SPECIFICATION

CAMERA, COPYING, LITHOGRAPHIC PROCESS, OVERHEAD, DARKROOM TYPE, 11 BY 14 INCHES (279 BY 356 mm)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of overhead, lithographic process copying camera; film size 11 by 14 inches (279 by 356 mm) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

DD-G-451	-	Glass, Plate, Sheet, Figured (Float, Flat, for Glazing, Corrugated Mirrors and Other Uses).
QQ-P-416	-	Plating, Cadmium (Electro-Deposited).

MILITARY

MIL-V-173	-	Varnish, Moisture and Fungus Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-C-3615	-	Cameras, Copying, Lithographic, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command (GL), Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this letter.

FSC 3610

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-150 - Photographic Lenses.
MIL-STD-461 - Electromagnetic Interference Characteristic Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristic Measurement Of.

DRAWINGS

ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

6-1-3250 - Camera, Copying, Lithographic Process Overhead Darkroom Type 11 by 14 inches; Assembly Complete.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70

(Copies of the National Electrical Code may be obtained from the National Fire Protection Association, International; 60 Batterymarch Street, Boston, MA 02110.)

NATIONAL BUREAU OF STANDARDS

Handbook H-28-Screw-Thread

Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for inspection and approval (see 4.3, 6.2 and 6.4).

3.2 Electrical code. All electrical wiring shall conform to NFPA No. 70.

3.2.1 Certification. Prior to the approval of the first article, or if none is submitted, prior to the approval of the first shipment, the contractor shall submit acceptable evidence that the equipment he proposes to supply under this specification meets the requirements of the National Electrical Code No. 70. Acceptable evidence of meeting these requirements shall be a certified test report from a nationally recognized independent testing laboratory approved by the contracting officer stating that the electrical wiring has been tested and conforms to the National Electrical Code No. 70.

3.3 Alternate components. Components offered as equivalent to components specified on the referenced drawings and hereinafter as specific manufacturer's part number or equal shall be functionally equal to the manufacturer's part number identified, and of equal or better quality. The incorporation and inclusion of such a component shall not require modification or change to any other specified component, and shall not reduce ease of operation or require relocation of any component, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article or if none is required prior to commencing production the contractor shall submit, for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of first article or production manufacture as applicable and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.4 Materials and components. Materials and components shall be as specified herein and as shown on the applicable drawings. Materials not definitely-specified shall be of the quality normally used by the manufacturer provided the completed item complies with all provisions of this specification.

3.4.1 Plate glass. Plate glass shall conform to type I, class 1, quality q3 of DD-G-451 and to the dimensions of item 5-17 of Drawing 6-1-3261. The thickness of the plate glass shall be a nominal 1/4 inch (6.4 mm).

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3.4.2 Varnish. Varnish shall conform to the requirements of MIL-V-173.

3.4.3 Rubber. The copy board pad shall be made from closed cell, firm density neoprene rubber conforming to Minor Rubber Company No. AB-432N or equivalent.

3.5 Performance. The camera shall reproduce black and white line and half-tone negatives. Half-tone negatives shall be made by use of a gray contact screen. Contact screen shall have not less than 65 grids per inch (2.6 grids per mm) and not more than 120 grids per inch (4.7 grids per mm).

3.5.1 Capacity. The film case shall accommodate standard sizes of cut photographic film from a minimum of 4 by 5 inches (102 by 127 mm) to a maximum of 11 by 14 inches (279 by 356 mm) for line and half-tone reproduction.

3.5.2 Range of reproduction. The reproduction ratio shall be from a reduction of 33-1/3 percent to an enlargement of 300 percent when equipped with the lens specified in 3.7.1.4.1. The image on the ground viewing glass shall be within a margin of error of plus or minus 1/64 inch (0.4 mm) of the test target for all ranges of reproduction when tested as specified in 4.4.6.

3.6 Design. The design shall provide positive locking of all members and parts. The camera shall be a horizontal stand overhead type with a stationary copyboard. The camera shall be a darkroom precision type. The movement of parts controlling the operation, focusing, or adjustment of the camera shall be positive, and have uniform resistance free from lost motion.

3.7 Construction. The construction of the camera shall conform to requirements specified herein and Drawing 6-1-3250, including the assembly and detail drawings referenced thereon.

3.7.1 Detail of components.

* 3.7.1.1 Copyboard assembly. The copyboard assembly shall accommodate copy up to 18 by 24 by 1/2 inch (457 by 610 by 12.7 mm) thick minimum and its center shall be coincident with the optical axis of the lens. When in the copying position, the copyboard shall be maintained in parallel alignment (tolerance ± 0.001 inch (0.025 mm)) with the camera focal plane when tested as specified in 4.5.2. The glass shall be as specified in 3.4.1. The neoprene rubber pad shall be as specified in 3.4.3. The rubber pad shall be permanently marked with horizontal and vertical center lines having graduations in one half inch (12.7 mm) increments to facilitate centering the copy. The copyboard assembly shall be fixed and rigidly attached to the monorail frame. The copyboard assembly shall be capable of pivoting 90° + 5°. The amount of pivoting shall be limited by a spring and 1/4 inch (6.4 mm) ladder chain secured to the monorail assembly.

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* 3.7.1.2 Lamp support assembly. The lamp support assembly shall provide for all positioning and adjusting of the lamp carrier arms. The lamps shall be secured in place without the use of hand tools. The lamp shall provide uniform coverage within a tolerance of plus or minus 15 percent of the entire copyboard when tested as specified in 4.5.4. The reflecting surfaces shall not blister or discolor when tested as specified in 4.5.4.

* 3.7.1.3 Lens carrier assembly. The lens carrier assembly shall be attached to the monorail assembly. The lens carrier assembly shall maintain parallel alignment (tolerance ± 0.001 inch (0.025 mm)) with the copyboard and film case along the entire length of track of the monorail carrier assembly when tested as specified in 4.5.2.

* 3.7.1.3.1 Lens. The camera shall be provided with a process lens conforming to type IV of MIL-STD-150. The lens shall be an apochromatic type, focal length of 10-3/4-inches (273 mm), a maximum relative aperature of f/9.5 and a minimum relative aperture of f/64.

* 3.7.1.3.2 Iris diaphragm. The iris diaphragm shall be manually operated and shall correlate the "f" values of the lens, to provide an equivalent transmission of light at various ratios of enlargement and reduction. The diaphragm control shall have graduated spacings between f-stop openings from f/9.5 to f/64.

* 3.7.1.4 Bellows. A bellows shall connect the lens case assembly to the film case assembly. The bellows shall provide the necessary extension and retraction to obtain the reproduction range specified in 3.5.2 without image obstruction in any position. The bellows shall be fabricated of molded rubber or impregnated rubberized materials, and shall have all corners reinforced. The bellows shall permit adjustment and movement without damage to the bellows material when tested as specified in 4.5.3. The bellows shall be supported along its length to prevent sag, if required. The entire bellows, including the joints between the sections, shall be light-tight when tested as specified in 4.4.3.

3.7.1.5 Film carrier assembly. The film carrier assembly shall be attached to the monorail assembly. The assembly shall maintain accurate alignment (tolerance ± .001 inch (0.025 mm)) along the length of track of the monorail carrier assembly when tested as specified in 4.5.2.

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3.7.1.5.1 Vacuum back. The vacuum back and zone control shall provide means to hold standard cut film sizes 4 by 5 through 11 by 14 inches (305 by 381 through 279 by 356 mm) and a gray contact screen 12 by 15 inches (305 by 381 mm). Visual means for centering film in each zone shall be provided. The film shall be held flat over its entire surface by the vacuum back without the use of supplementary means when tested as specified in 4.5.5. The surface of the vacuum back directly under the film shall cause no distortion, when tested as specified in 4.5.5. The film shall not lift from the surface of the vacuum back when tested for film adherence as specified in 4.5.5.

3.7.1.5.2 Ground viewing glass. The ground viewing glass shall be fabricated of glass specified in 3.4.1. One surface of the glass shall be a uniformly grained surface produced by a fine acid, or mechanically capable of resolving not less than 20 lines per millimeter in a projected image. The glass shall have permanent markings to permit visual determination of the position of the image in relation to the film position on the vacuum back. The grained surface shall have a 1-inch (25.4 mm) clear square or circle in its center and clear diagonals. The edges of the glass shall be ground square.

3.7.1.6 Vacuum system. The vacuum system shall operate without objectionable noise or vibration.

3.7.1.7 Focusing. The percent gallery, 10-3/4 lens, and focusing scale shall be permanently engraved and shall be of the percentage type. The camera shall be in accurate focus when the calibrated scales are set and tested as specified in 4.5.6.

3.7.2 Controls.

3.7.2.1 Copyboard illumination control. A switch and a power control unit shall be provided to control the operation and intensity of the copyboard illuminating lamps.

3.7.2.2 Shutter control. The camera shall be equipped with an electric solenoid shutter control. The shutter shall be designed to open or close manually or automatically. Illumination of the copyboard shall be controlled either manually or automatically at a predetermined time setting. The 5-minute time setting device shall be calibrated in minutes and seconds, and shall be capable of repeating the same cycle, or of being reset for other time cycles.

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3.7.3 Electrical components. The electrical components shall be designed to operate on 120 volt, 60 hertz, single phase power source. All wiring shall conform to the National Electrical Code No. 70 (see 3.2). All wiring shall be color coded. An 8 by 10-1/2 inch (203 by 381 mm) laminated plastic inclosed or an etched metal wiring diagram, detailing color coded wired, and identifying switches and controls by name and part number shall be furnished.

3.7.3.1 Electromagnetic compatibility. When specified (see 6.2), the equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.5.7).

3.7.3.2 Fungus and water resistance. The electrical circuity, including all components and connections shall be protected from the effect of water and fungus growth by an overall treatment with the varnish specified in 3.4.2. Components or circuit elements inherently fungus and water resistant, hermetically sealed, or affected by the varnish coating need not be treated. All surfaces to be treated shall be cleaned and prepared prior to application of the varnish. The varnish shall be applied by spray, brush or a combination of both to give a uniform dry film thickness.

3.7.4 Fastening devices. Screws, nuts and washers shall be in the types and sizes specified on the drawing . Carbon steel fasteners shall be finished as specified in 3.9.2. All screw threads shall conform to the National Bureau of Standards Handbook H-28.

3.8 Lubrication. Oil holes, grease cups, or other means of lubrication shall be provided for lubricating all parts subject to wear. Parts that must not be lubricated, such as the lens diaphragm, shall be marked with a "DO NOT LUBRICATE" warning.

3.9 Finish.

3.9.1 Painting. All exterior and interior surfaces of the camera that are normally painted shall be treated and painted in accordance with the manufacturer's commercial finish. The color shall be blue gray conforming to No. 36231 of FED-STD-595.

3.9.2 Plating. All carbon steel fasteners screws, nuts and washers required to fabricate the camera shall be cadmium plated in accordance with class 3, type II of QQ-P-416.

3.10 Marking for identification. The camera shall be marked for identification in accordance with MIL-STD-130 and shall include the year of manufacture.

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3.11 Test target. The camera shall be provided with a test target for periodically checking the accuracy and calibration of the camera. The test target may be made of a polyester plastic or any stable material in the form of a flat scale or a ruled grid. The flat scale shall be accurately calibrated in 1/64 inch (0.4 mm) increments in black on a white background and shall be capable of being located in any position on the copy board. A protective container shall be provided for the test target.

3.12 Tools. The camera shall be provided with one complete set of all tools necessary to assemble, adjust and dismantle the camera.

3.13 Workmanship. The end item shall be properly assembled. There shall be no loose, misplaced or missing components. All fastening devices shall be drawn tight. All functioning components shall operate without binding or excessive looseness. All rough and sharp edges shall be removed. Electrical wiring shall be secure at terminals with adequate slack to prevent strain.

3.13.1 Welding. The surface of parts to be welded shall be thoroughly cleaned. Welds shall be continuously and uniformly run and free of voids, scabs and blisters. All welding scale and flux shall be removed after welding is accomplished. All projections and rough edges shall be dressed smooth. Spot welds, when utilized, shall be sound and smooth, with no pits or flashes.

3.13.2 Castings. Castings shall be free from cracks, porosity, blow holes, warp or other casting imperfections.

3.13.3 Enamel. The enamel shall level out to a uniform dry film without objectionable orange peel, wrinkles, drops, streaks or areas of no film.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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* 4.3 First article inspection. When a first article is required, the camera shall be examined for defects in table I, the dimensions specified, and tested in accordance with 4.5.1 through 4.5.7, as applicable. The presence of any defect, any dimension not within specified tolerance or failure of any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 End item inspection.

4.4.2.1 Visual examination. Each camera shall be examined for defects in table I. The presence of any defect shall be cause for rejection of the camera.

TABLE I. Visual examination for defects

Examine	Defect
Copyboard assembly	Long dimension not in vertical plane
	Pad missing, not of material specified, graduated vertical and horizontal centerlines missing
	Glass scratched or cracked, locking screws missing
	Adjusting screws missing or broken
	Locking devices missing or broken
	Retaining chain missing or of improper length, spring missing
Monorail assembly	Rails nicked, gouged or not machined smooth
	Wire reels missing or broken
	Rubber grommet missing from copy board frame support
	Copyboard release knob missing or broken
Lens case and camera back carrier assembly	Not supported by roller bearings
	Adjusting screws missing
	Cases and carriers out of alignment with one another
	Holes not drilled on both left and right carriers of each case
	Carrier assembly base not flush with each case

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TABLE I. Visual examination for defects (cont'd)

Examine	Defect
Lamp assembly	Light arms adjusting knobs missing
	Quadrant plate not secure, angle marks missing
	Longitudinal arms not laterally adjustable around quadrant plate or in and out
	Longitudinal slot missing or not long enough in the lateral arms
	Vertical arms not secure to the lateral arms
	Locking screw missing at the lateral-vertical connecting point
	Wiring not proper within the assembly
	Reflectors loose, not as specified
	Reflector retaining collar missing
	Variable light source control missing, broken, not as specified
Lens and lens case assembly	Lens board screen missing, not secure
	Lens cap missing, lens scratched, incorrect focal length, stained, not apochromatic, no waterhouse slot
	Iris diaphragm scale graduations missing or illegible
	Iris diaphragm scale indicator missing, broken, not type specified
	Timer not mounted on operator's side, not type specified, damaged
	Solenoid and shutter not as specified, missing or damaged
	Lens storage box without cover and latch
Bellows assembly	Not fabricated of molded rubber or impregnated rubberized material
	Corners not reinforced
	Cracked or otherwise damaged
	Not properly supported

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TABLE I. Visual examination for defects (cont'd)

Examine	Defect
Film case carrier assembly	Does not pivot about horizontal axis
	Locking mechanism missing, not as specified
	Zone control scale decal missing or illegible
	Zone marking on vacuum back missing
	Vacuum back frame missing, out of alignment, retaining screws missing
	Adhesive back missing
	Zone control knob missing, loose or broken
	Ground viewing glass cracked, scratched or or stained; markings missing or illegible; not housed in metal frame
	Light tight baffle missing, not as specified
	Screws missing on hinges, binds when pivoting
	Screen not type or size specified, grid size not as specified
Vacuum system	Pump not as specified
	Motor not as specified
	Pump and motor not mounted as integral unit
	Hose line missing, frayed, cracked or otherwise damaged
	Control switch missing
Focusing assembly	Proportional scale markings missing or illegible
	Scales not identified in percentages
	Hairline indicator missing or broken
	Focusing knob missing, broken, or not as specified
	Focusing scale cover missing, not correct size, not material specified
Electrical components	Wires broken, frayed or otherwise impaired, not type specified
	Wiring diagram missing
	Terminal blocks not as specified
	Fungus and water resistance missing where required

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TABLE I. Visual examination for defects (cont'd)

Examine	Defect
Welding	Weld not smooth and uniform Weld splatter, blister, or pock mark Weld incomplete, cracked, porous or not fused
Bolts, nuts, and screws	Loose, missing, broken threads, stripped threads or otherwise damaged
Finish	Not uniform, not as specified
Instruction plates	Missing
Test target	Missing, or not as specified; scratched or otherwise damaged
Identification marking	Missing, illegible

* 4.4.2.2 Dimensional examination. Examination shall be made of each camera for compliance with dimensions specified. Any dimension not within specified tolerance shall be cause for rejection of the item.

* 4.4.3 End item testing. Each camera shall be tested as specified in 4.5.1 through 4.5.6 and failure of any test shall be cause for rejection of the item. When applicable, the test as specified in 4.5.7 shall be performed on one representative production unit and failure of the test shall be cause for rejection of the item.

4.4.4 Code and standards compliance. Proof of compliance with NFPA No. 70 (see 3.2.1) shall be made available to the Government representative.

4.4.5 Packaging inspection. Examination for preservation, packaging, packing and marking shall be in accordance with quality assurance provisions of MIL-C-3615.

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4.5 Tests.

4.5.1 Operational tests. Actual operating conditions for the camera shall be assimilated in order to check all functional and electrical components to insure their performance as intended. These components will include the solenoid, shutter, timer, light package, and vacuum pump. The camera shall be tested for proper operation in accordance with 4.5.2 through 4.5.7, as applicable. Testing shall be conducted in a room at a temperature range of 68 to 75F (20 to 23.8 C) and at a 45 to 50 percent relative humidity.

4.5.2 Parallelism test. The parallelism test of the lens case, film case and copyboard shall be made by means of a telescoping trammel rod with a dial indicator (calibrated in 0.001 inch (0.025 mm) increments) attached at one end. Parallelism of these components shall be checked at three different ratios of reproduction and enlargements covering the reproduction range of the lens to determine compliance with 3.7.1.1, 3.7.1.3 and 3.7.1.5. Any nonconformance shall constitute failure of this test.

4.5.3 Material adjustment and light-tightness. With the lens shutter closed, the lens case assembly and the film case assembly shall be moved to their respective 33-1/3 percent positions. Such positioning will extend the bellows to their maximum extension. The copyboard illuminating lamps shall then be directed towards the lens and bellows, and a black drop cloth shall be placed over the film case assembly. All room lights shall be turned off for five minutes to allow the eyes of the inspector to become dark-adapted. The copyboard illuminating lamps shall then be turned on and the interior of the entire assembly shall be examined for light-tightness to determine compliance with 3.7.1.5. Examination shall be conducted under cover of the black drop cloth. Any nonconformance shall constitute failure of this test.

4.5.4 Copyboard illumination. A sheet of white paper 18 by 24 inches (457 by 610 mm) shall be subdivided into twenty-five equal rectangles. The sheet shall be placed on the copyboard. The lamp carrier assembly shall be positioned as far away from the copyboard as possible. The lamps shall be adjusted so that maximum uniform coverage of the copyboard is obtained. After adjusting the lamps, the copyboard shall be illuminated and all room lighting shall be turned off. Light meter readings shall be taken at the approximate center of each of the twenty-five rectangles to determine compliance with 3.7.1.3. At the completion of test, the lamp reflecting surfaces shall be examined to determine compliance with 3.7.1.3. Any nonconformance shall constitute failure of this test.

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4.5.5 Vacuum back. The vacuum back shall be tested for film adherence and distortion by using six separate sheets of film ranging in sizes from 4 by 5 inches to 11 by 14 inches (102 by 127 mm to 279 by 356 mm). Each sheet of film shall be tested separately. The film shall be placed on the vacuum back which shall be in the horizontal position and the vacuum pump started. When the pump has reached its rated vacuum, a light shall be placed incident to the film surface and moved over the entire surface. The film shall be examined for flatness and for denting and pulling out of place to determine compliance with 3.7.1.5.1. A corner of the film shall be raised in order to apply a force to the film in the lateral direction. The applied force shall cause the film to move slowly over the surface of the vacuum back to determine compliance with 3.7.1.5.1. Any nonconformance shall constitute failure of this test.

4.5.6 Performance test. The test target shall be placed centrally on the copyboard. An 18 inch rule calibrated in 1/64 inch (0.4 mm) increments shall be placed on either side of the vertical centerline of the copyboard. The rule increments shall be black on a white background. When the copyboard is in copying position, the lights shall be placed 2 feet (610 mm) from the center of the copyboard at an angle of 45 degrees. The light source control panel shall be set for 3200 degrees Kelvin. With the automatic timer on manual control the camera focusing scales shall be set at a 1:1 ratio and the lens aperture set at f/22. All room lighting shall be turned off and the copyboard lamps turned on. A 5 power magnifying glass shall be placed on the ground glass to check the sharpness of the image. The resultant image on the ground viewing glass shall be measured with a rule and compared to test target to determine compliance with 3.5.2. The increment position of the focusing scales shall be checked through the gunsight indicator. Both scales shall measure 100 percent to determine compliance with 3.7.1.8. Repeat this procedure for reproductions of 50 percent, 33-1/3 percent, and for enlargement of 200 and 300 percent. Exposures shall be made at each of the enlargement and reduction ratios and the resultant size of the negative image measured to determine compliance with 3.5.2. The film length and width shall be measured before exposure and after development to determine if resultant image or film is distorted. Any nonconformance shall constitute failure of this test.

* 4.5.7 Electromagnetic compatibility. When electromagnetic compatibility is required, the unit shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging, for level A, B or C, shall be in accordance with MIL-C-3615 (see 6.2).

6. NOTES

6.1 Intended use. The camera covered by this specification is intended for mobile installation for printing plant special warface semi-trailer mount and use in the production of black and white line and half-tone negatives for the lithographic process.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3 and 6.4).
(c) When electromagnetic compatibility is required (see 3.7.3.1).
(d) Selection of applicable levels of preservation, packaging and packing (see 5.1).

* 6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically cited in the contract in accordance with the applicable regulations of the procuring activity.

6.3 Engineering conference. Within five days after award of contract unless otherwise specified, and prior to any work on preproduction sample including ordering of materials, the contracting officer shall arrange an engineering conference to discuss any questions concerning the details of Government drawings, instructions and technical requirements of this specification. The conference is to assist the contractor in interpretation of the specification, and agreements made shall neither obligate the Government to accept equipment that is not in accordance with this specification nor relieve the contractor of his obligation to supply the equipment specified.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

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6.5 Changes from previous issue. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, deletions, modifications, corrections) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

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Air Force - 99

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